

“AI-Powered Disease Prevention & Outbreak Intelligence Platform”

Predict Early → Prevent Faster → Act Smarter

Team name : 404 THINKERS

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Problem Statement: current Public Health Systems Are Reactive

Current workflow:

People get sick

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Hospitals overflow

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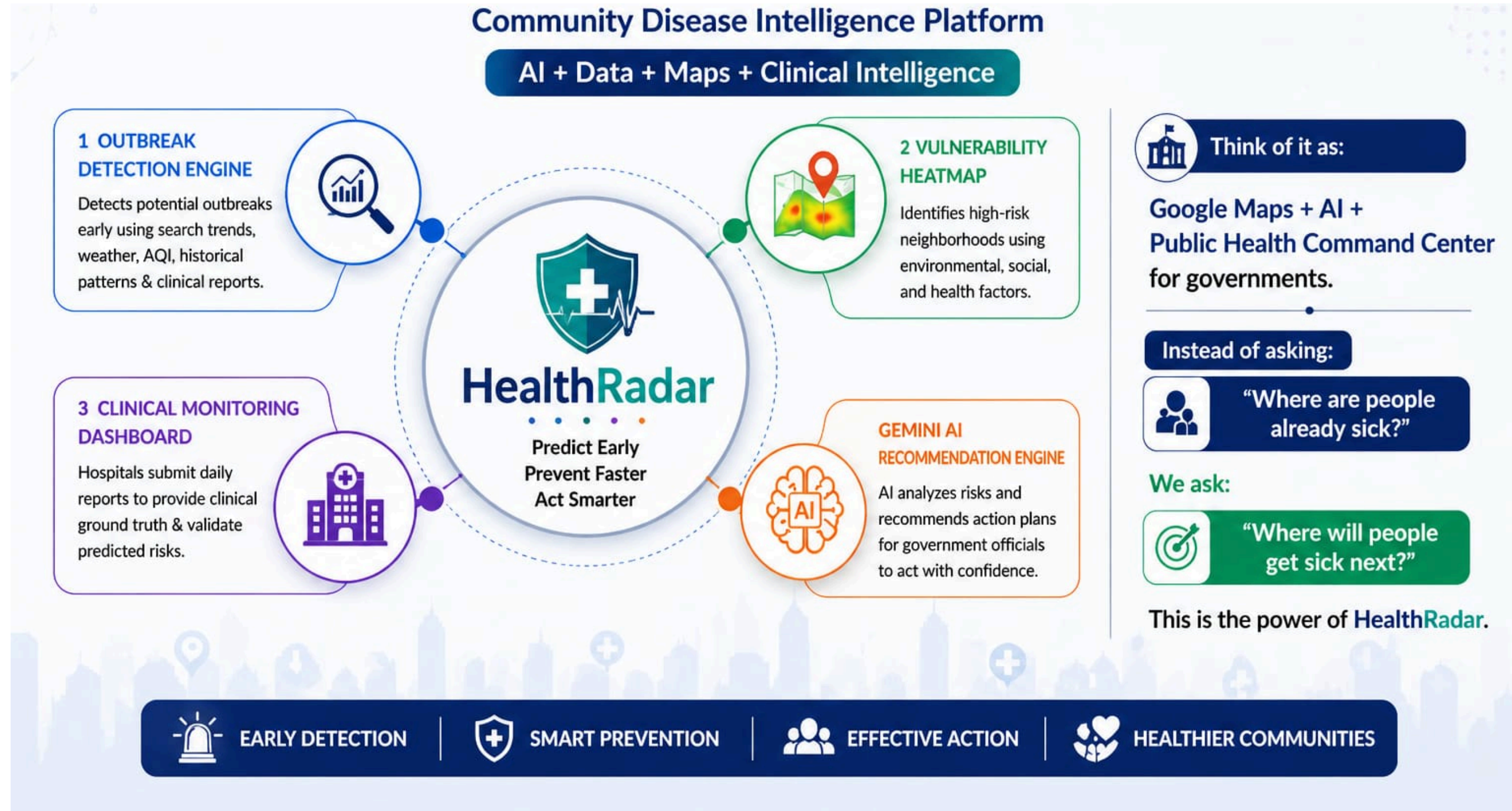
Government reacts

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Too late

Problems:

- delayed outbreak response
 - preventable deaths
 - poor resource allocation
 - blind vaccine deployment
 - high-risk communities suffer more
- Strong emotional impact.



Core Features:

Our 4 Major Solution Layers

1. Outbreak Detection Engine

Predict outbreaks early

2. Vulnerability Heatmap

Identify high-risk neighborhoods

3. Clinical Monitoring Dashboard

Hospital reporting + validation

4. Gemini AI Recommendation Engine

Government action suggestions (If SEIR removed, do not include)

This slide must look clean.



System Architecture:

hospital_master.csv

+

clinical_reports.csv

+

weather_data.csv

+

aqi_data.csv

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Python Data Engine

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Gemini API

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AI Recommendations

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Streamlit Dashboard

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Government Decision Support

Technology Stack

Frontend:

- Streamlit
- Plotly
- Folium

Backend:

- Python

Database:

- Firebase
- Google Sheets

Development:

- VS Code
- GitHub
- Antigravity

AI Layer:

- Gemini API

APIs:

- Weather Data
- AQI Data
- Google Maps Platform

Live Dashboard Screens:

- Outbreak Dashboard
- Vulnerability Heatmap
- Clinical Monitoring Dashboard
- AI Recommendations Screen

Gemini AI Intelligence

Example:

Location: KR Puram

Disease: Dengue

Severity Score: 87/100

Risk Level: HIGH

Gemini Recommendations:

- mosquito fogging
- temporary fever camps
- reserve 30 hospital beds
- awareness campaigns near schools
- local testing camps

Real-World Impact

Healthradar helps governments:

- prevent outbreaks early
- reduce preventable deaths
- optimize vaccines/resources
- improve hospital readiness
- protect vulnerable communities
- move from reactive → preventive healthcare

Future Scope

- real-time Google Trends
- live AQI + weather APIs
- district-level government deployment
- multilingual citizen alerts
- SMS outbreak warnings
- state-wide outbreak monitoring
- national disease intelligence platform shows scalability.

THANK YOU